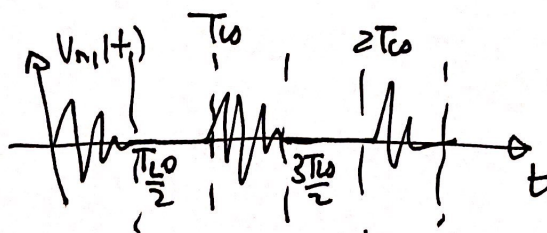


Homework 5: Chapter 6, Mixers:

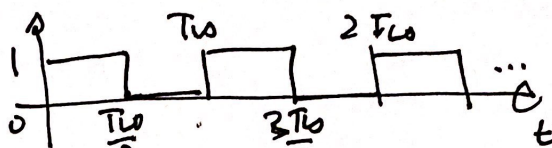
problem 6.4:

$$V_{n1}(t) = V_{n1,LPF}(t) \cdot S(t)$$



$$S(t) = \prod\left(\frac{t}{\frac{T_{Lo}}{2}} - \frac{1}{2}\right) * \sum_{k=-\infty}^{+\infty} \delta(t - kT_{Lo})$$

$$S(f) = F[S(t)]$$



$$S(f) = \frac{1}{T_{Lo}} \sum_{k=-\infty}^{+\infty} \delta(f - kf_{Lo}) \cdot \left[\frac{1 - e^{-j\omega T_{Lo}/2}}{j\omega} \right]$$

$$= \frac{1}{T_{Lo}} \sum_{k=-\infty}^{+\infty} \delta(f - kf_{Lo}) \cdot \frac{\text{Si}\left(\frac{\omega T_{Lo}}{4}\right)}{(\omega/2)} e^{-j\omega \frac{T_{Lo}}{4}}$$

$$= \frac{1}{2} \sum_{k=-\infty}^{+\infty} \delta(f - kf_{Lo}) \cdot \text{Sinc}\left(\frac{\omega T_{Lo}}{4}\right) e^{-j\omega \frac{T_{Lo}}{4}}$$

$$\overline{V_{n1}^2}(f) = \overline{V_{n1,LPF}^2}(f) * |S(f)|^2$$

$$= \frac{2kTR_1}{1 + (2\pi f R_1 C)^2} * \frac{1}{4} \sum_{k=-\infty}^{+\infty} \text{Sinc}^2\left(\frac{\omega T_{Lo} k}{4}\right) \delta(f - kf_{Lo})$$

two-sided noise spectrum

$$\overline{V_{n1}^2}(f) = \frac{2kTR_1}{1 + (2\pi f R_1 C)^2} * \frac{1}{4} \sum_{k=-\infty}^{+\infty} \text{Sinc}^2\left(k \frac{\pi}{2}\right) \delta(f - kf_{Lo})$$

$k=0 \rightarrow$ this low frequency component is excluded as it is correlated with $V_{n2}(t)$

$$k=\pm 1 \rightarrow \text{Sinc}^2\left(\frac{\pi}{2}\right) = \frac{4}{\pi^2} \times 2 \quad k=\pm 2, \pm 3, \dots \rightarrow \text{Sinc}^2\left(\frac{k\pi}{2}\right) = 0$$

$$k=\pm 3 \rightarrow \text{Sinc}^2\left(\frac{3\pi}{2}\right) = \frac{4}{9\pi^2} \times 2$$

$$k=\pm 5 \rightarrow \text{Sinc}^2\left(\frac{5\pi}{2}\right) = \frac{4}{25\pi^2} \times 2$$

Since $\frac{1}{2\pi R_1 C_1}$ rarely exceeds $3f_{LO}$, noise aliasing components $\frac{2}{9\pi^2}$ for $|k| > 3$ are ignored

$$\bar{U}_{n_1}^2(f) = \frac{2kTR_1}{1 + (2\pi f R_1 C_1)^2} * \left[2 * \left(\frac{\delta(f-f_{LO})}{\pi^2} + \frac{\delta(f-3f_{LO})}{9\pi^2} \right) \right]$$

$$\bar{U}_{n_1}^2(f) = \frac{2kTR_1}{1 + (2\pi f_{LO} R_1 C_1)^2} \left(\frac{2}{\pi^2} \right) + \frac{2kTR_1}{1 + (2\pi (3f_{LO}) R_1 C_1)^2} \left(\frac{2}{9\pi^2} \right)$$

→ At low input frequencies ($f \ll f_{LO}$)

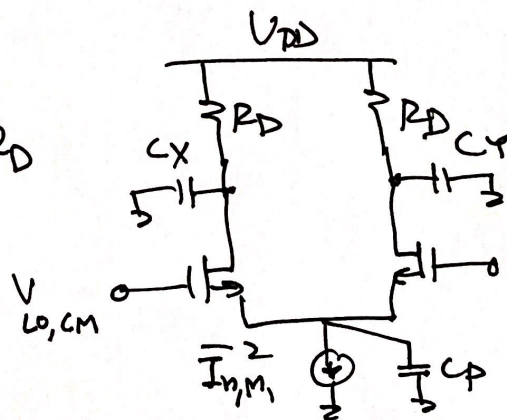
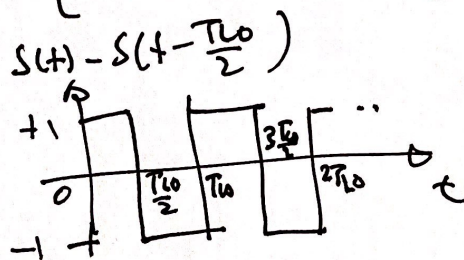
$$\frac{2kTR_1}{1 + 2\pi f_{LO} R_1 C_1} \approx 2kTR_1, \quad \frac{2kTR_1}{1 + 2\pi (3f_{LO}) R_1 C_1} \approx 2kTR_1$$

$$\begin{aligned} \bar{U}_{n_1}^2(f) &\approx (2kTR_1) (2) \left[\frac{1}{\pi^2} + \frac{1}{9\pi^2} \right] \\ \bar{U}_{n_1}^2(f) &\approx (0.226) (2kTR_1) \end{aligned}$$

Problem 6.37

$$\bar{U}_{n_{10}}^2 = \left[\bar{I}_{n_{M_1}}(t) \cdot \left[S_D(t) - S\left(t - \frac{T_{LO}}{2}\right) \right] \right]^2 R_D$$

Noise Voltage at the output due to M_1



$$\bar{I}_{n_{M_1}}^2 = 4kT\gamma g_{m1}$$

$$\bar{U}_{n_{10}}^2(f) = \left[\bar{I}_{n_{M_1}}^2(f) * |S_D(f)|^2 \right] R_D^2$$

$$S_D(f) = \sum_{\substack{n=-\infty \\ \text{odd}}}^{+\infty} \frac{2}{\pi|n|} \delta(f - n f_{LO}) \rightarrow |S_D(f)|^2 = \sum_{n_{\text{odd}}} \frac{4}{\pi^2 |n|^2} \delta(f - n f_{LO})$$

$$\bar{U}_{n,0}^2(f) = \left[\bar{I}_{m,M_1}^2(f) * \sum_{n, \text{odd}} \frac{4}{\pi^2 n^2} \delta(f - n f_{LO}) \right] R_D^2$$

$$\begin{aligned} \bar{U}_{n,0}^2(f) &= \left[\sum_{n, \text{odd}} \frac{4}{\pi^2 n^2} \bar{I}_{m,M}^2(f - n f_{LO}) \right] R_D^2 \\ &= \left[\left(\frac{4}{\pi^2} \times 4KT\gamma g_m \right) \left(\sum_{\substack{n, \text{odd} \\ n \rightarrow \infty}} \frac{1}{n^2} \right) \right] R_D^2 \\ &= \frac{4KT\gamma}{g_m} \times g_m^2 R_D^2 \times \frac{4}{\pi^2} \left(2 \sum_{\substack{n=1 \\ n, \text{odd}}}^{\infty} \frac{1}{n^2} \right) \end{aligned}$$

$$2 \sum_{n, \text{odd}} \frac{1}{n^2} = 2 \left[1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] = 2 \times \frac{\pi^2}{8} = \frac{\pi^2}{4}$$

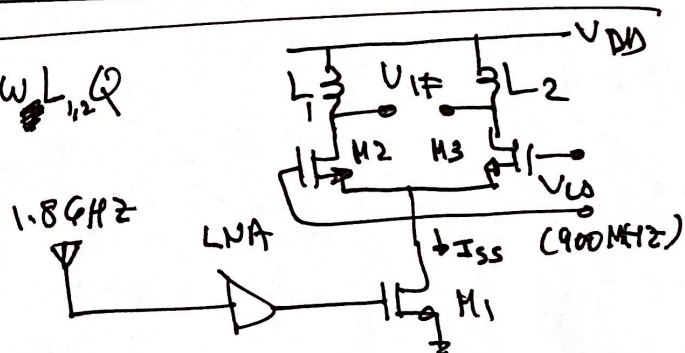
$$\rightarrow \bar{U}_{np}(f) = \frac{4KT\gamma}{g_m} \underbrace{\left(\frac{2g_m R_D}{\pi} \right)^2}_{CG} \cdot \frac{\pi^2}{4}$$

$$\rightarrow \boxed{\bar{U}_{n,0}^2(f) = \frac{\pi^2}{4} \times (4KT\gamma/g_m) \times CG^2}$$

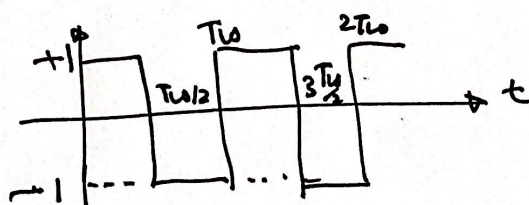
Problem 6.9 $\rightarrow L_1, L_2$ resonate @ IF, $R_p = \omega L_{1,2} Q$

(a) M_2, M_3 switching abruptly

LO-IF feedthrough (900 MHz o/p in the absence of RF signal):



$$\rightarrow V_{IF}(t) = \left[S(t) - S\left(t - \frac{T_{LO}}{2}\right) \right] \times \overset{\text{constant (to find LO-IF leakage)}}{I_{SS} \times R_p}$$



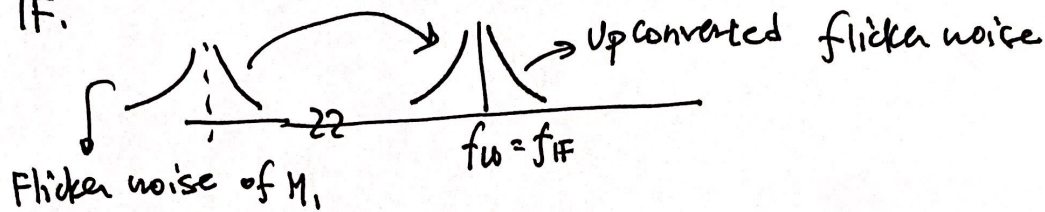
$$S(t) - S\left(t - \frac{T_{LO}}{2}\right) = \sum_{n, \text{odd}} \frac{4}{n\pi} \left[\cos(n\omega_{LO} t) \right]$$

$$V_{IF}(t) = \left[\frac{4}{\pi} \cos(\omega_{LO} t) + \dots \right] I_{SS} R_p \rightarrow \text{LO-IF leakage} = \frac{4}{\pi} I_{SS} R_p$$

$$\text{or } \boxed{\text{LO-IF leakage} = \frac{4}{\pi} I_{SS} (\omega_{LO} L Q)}$$

(b) The flicker noise of M_1 is upconverted to $f_{LO}, 3f_{LO}, 5f_{LO}, \dots$ at the IF, but since $f_{IF} = 1.8^{GHz} - f_{LO} = 1.8^{GHz} - 0.9^{GHz} = 0.9^{GHz}$

$f_{IF} = f_{LO} \rightarrow$ the flicker noise of M_1 is directly upconverted to IF.



Problem 6.16

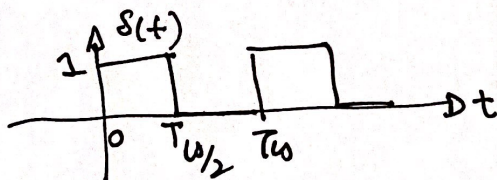
• Dual-gate mixer:

(a) conversion gain

M_2 doesn't enter triode

$\rightarrow g_{m2}$

$$V_{IF}(t) = \frac{2}{R_{F(on)}} \times S(t) \times R_D \times \frac{1}{2} V_{RF}(t)$$



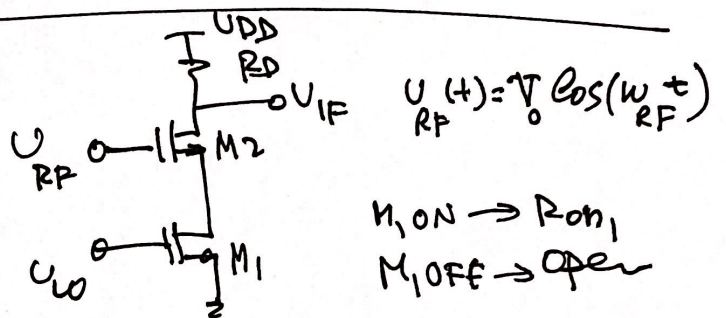
$$\rightarrow S(t) = \frac{1}{2} + \left(\frac{2}{\pi} \cos(\omega_{LO} t) + \frac{2}{3\pi} \cos(3\omega_{LO} t) + \dots \right) = \frac{1}{2} + \sum_{n=1, \text{ odd}}^{\infty} \frac{2}{\pi n} \cos(n\omega_{LO} t)$$

$$V_{IF}(t) = \frac{g_{m2} R_D}{1 + g_{m2} R_{on1}} \cdot V_0 \cos(\omega_{RF} t) \cdot \left[\frac{1}{2} + \frac{2}{\pi} \cos(\omega_{LO} t) + \frac{2}{3\pi} \cos(3\omega_{LO} t) + \dots \right]$$

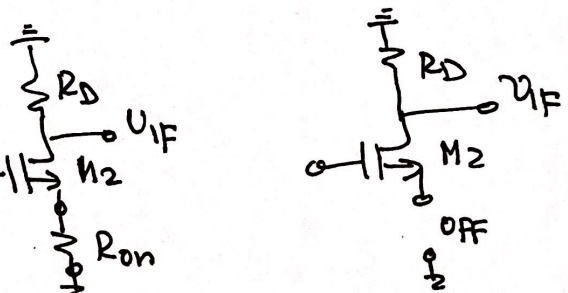
$\rightarrow$ Considering only the fundamental component of LO

$$V_{IF}(t) = \left[\frac{g_{m2} R_D}{1 + g_{m2} R_{on1}} \cdot \frac{V_0}{2} \cdot \frac{2}{\pi} \right] \cos(\omega_{IF} t)$$

$$CG = \frac{g_{m2} R_D V_0}{1 + g_{m2} R_{on1}} \cdot \frac{2}{\pi} \cdot \frac{1}{2} / V_0 \rightarrow \boxed{CG = \frac{g_{m2} R_D}{\pi (1 + g_{m2} R_{on1})}}$$



$M_1 \text{ ON} \rightarrow R_{on1}$
 $M_1 \text{ OFF} \rightarrow \text{open}$



$$R_{F(on)} = \left(\frac{g_{m2}}{1 + g_{m2} R_{on1}} \right) V_{RF}(t)$$

(b) If R_{on1} is very small, determine $1/P_2$ ($V_{ov2} = V_{GS0} - V_{TR}$) ⁵

$$i_D = \left(\frac{K'_n}{2}\right)\left(\frac{W}{L}\right) \left[V_{RF} + V_{ov2}\right]^2 = \left(\frac{K'_n}{2}\right)\left(\frac{W}{L}\right) \left[V_{RF}^2 + 2V_{ov2}V_{RF} + V_{ov2}^2\right]$$

$$i_D = \alpha_0 + \alpha_1 V_{RF} + \alpha_2 V_{RF}^2 \quad \left\{ \begin{array}{l} \alpha_1 = \left(\frac{K'_n}{2}\right)\left(\frac{W}{L}\right)(2V_{ov2}) = g_{m2} \\ \alpha_2 = \left(\frac{K'_n}{2}\right)\left(\frac{W}{L}\right) \end{array} \right.$$

$$1/P_{2,M1} = \frac{\alpha_1}{\alpha_2} = \frac{\left(\frac{K'_n}{2}\right)\left(\frac{W}{L}\right)V_{ov2}}{\left(\frac{K'_n}{2}\right)\left(\frac{W}{L}\right)} \rightarrow \{ 1/P_{2,M2} = 2V_{ov2}$$

$$1/P_{2,mixer} = 1/P_{2,M1} \times \left(\frac{GG}{RF-IF \text{ leakage}}\right) = (2V_{ov2}) \times \frac{\frac{1}{\pi} \cdot \frac{g_{m2}R_D}{1+g_{m2}R_{on1}}}{\frac{1}{2} \cdot \frac{g_{m2}R_D}{1+g_{m2}R_{on1}}}$$

$$\boxed{1/P_{2,mixer} = \frac{2}{\pi} \cdot (2V_{ov2})}$$

problem 6.17: (v_o, g_{mb} are negligible)

LO $\rightarrow$ abrupt edges with 50% duty cycle

(a)

$$V_{o1} = I_{SS} \times S(t) \times R_D$$

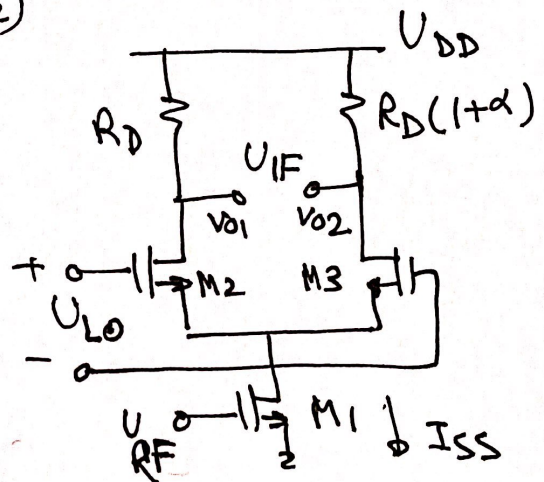
$$V_{o2} = I_{SS} \times S(t - \frac{T_{lo}}{2}) \times [R_D(1+\alpha)]$$

$$S(t) = \frac{1}{2} + \sum_{\substack{n=1 \\ n, \text{ odd}}}^{\infty} \frac{2}{n\pi} \cos(n\omega_{LO}t)$$

$$S(t - \frac{T_{lo}}{2}) = \frac{1}{2} + \sum_{\substack{n=1 \\ n, \text{ odd}}}^{\infty} -\frac{2}{n\pi} \cos(n\omega_{LO}t)$$

$$V_{o,DC} = V_{o1,DC} - V_{o2,DC} = \frac{I_{SS}R_D}{2} - \frac{I_{SS}R_D}{2} - \frac{\alpha I_{SS}R_D}{2}$$

$$\rightarrow \boxed{V_{o,DC} = -\frac{\alpha I_{SS}R_D}{2}}$$



$$\rightarrow V_{o1,DC} = \frac{I_{SS}R_D}{2}$$

$$\rightarrow V_{o2,DC} = \frac{I_{SS}(1+\alpha)R_D}{2}$$

(b) I_{P_2} in terms of V_{ov1} ; $I_{RFM_1} = \frac{K'_n}{2} \left(\frac{W}{L} \right) (V_{RF} + V_{ov2})^2$

$$I_{RFM_1} = \underbrace{g_{m1} V_{RF}}_{K'_n \left(\frac{W}{L} \right) V_{ov1}} + \frac{K'_n}{2} \left(\frac{W}{L} \right) V_{RF}^2 + \underbrace{I_{SS}}_{\frac{K'_n}{2} \left(\frac{W}{L} \right) V_{ov2}^2}$$

$$\rightarrow I_{P2, M_1} = 2V_{ov1} = \frac{\alpha_2}{\alpha_1}$$

$$I_{P2, mixer} = I_{P2, M_1} \cdot \left| \frac{\text{Conversion Gain}}{\text{RF-IF Leakage}} \right|$$

$$I_{P2, mixer} = 2V_{ov2} \cdot \frac{\frac{2}{\pi} R_D}{\alpha \frac{R_D}{2}} = \frac{8V_{ov2}}{\pi \alpha}$$

$$\rightarrow \boxed{I_{P2, mixer} = \frac{8V_{ov2}}{\pi \alpha}}$$